Public Policy 2026-2027

Lecture 5: Market Failures

Thursday, October 28, 2026
Stockholm School of Economics in Riga

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Keeping Efficiency and Equity Separate

A lesson learnt from neo-classical economics

First Welfare Theorem

Under ideal assumptions, competitive markets will always lead to a Pareto-efficient outcome.

Efficiency first!

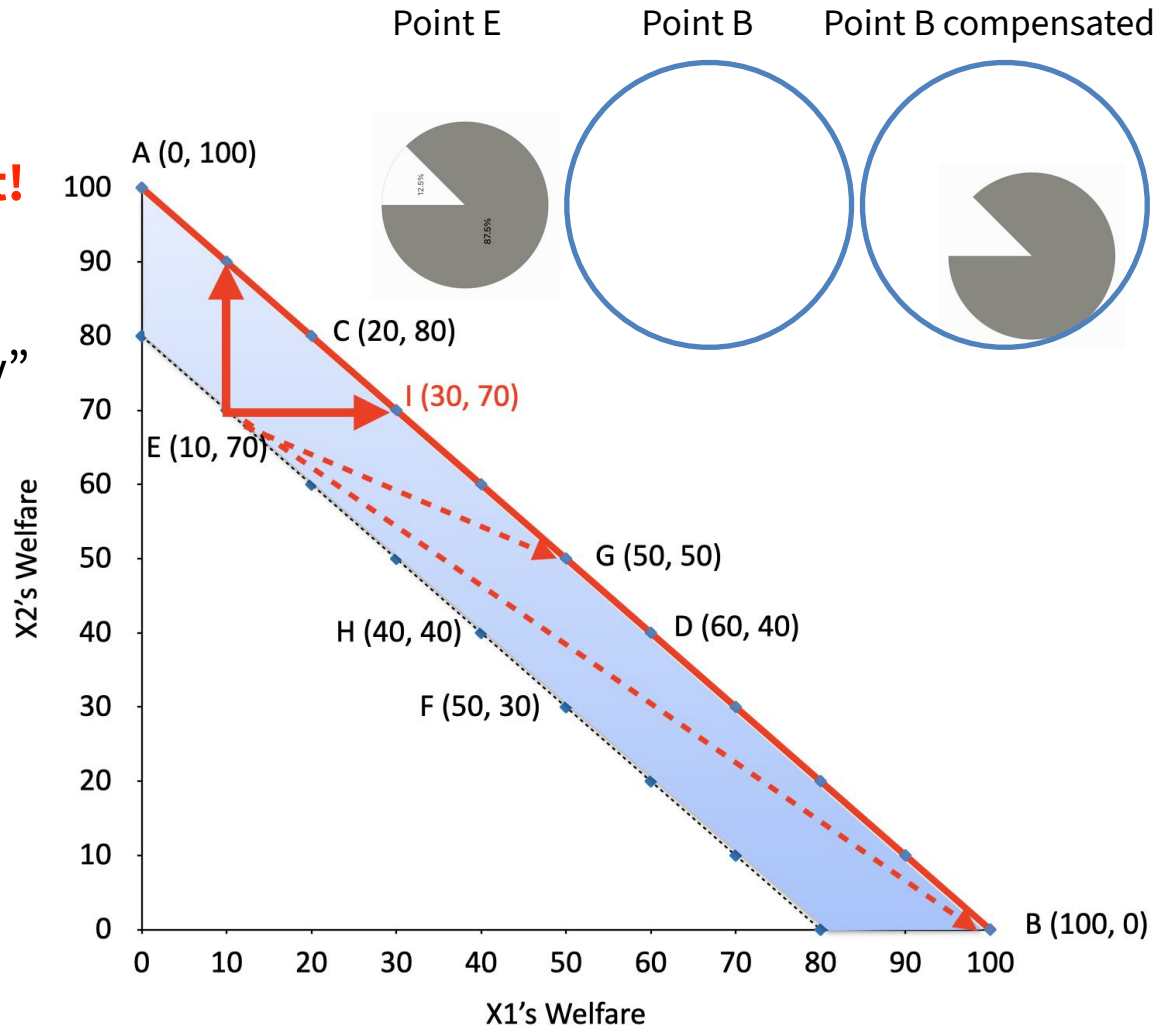
"Choose the large-pie policy"

Second Welfare Theorem

Society can attain any Pareto-efficient outcome by redistributing initial wealth among individuals and then allowing them to freely trade.

Redistribution second!

"Use transfers to reach the distribution you want"

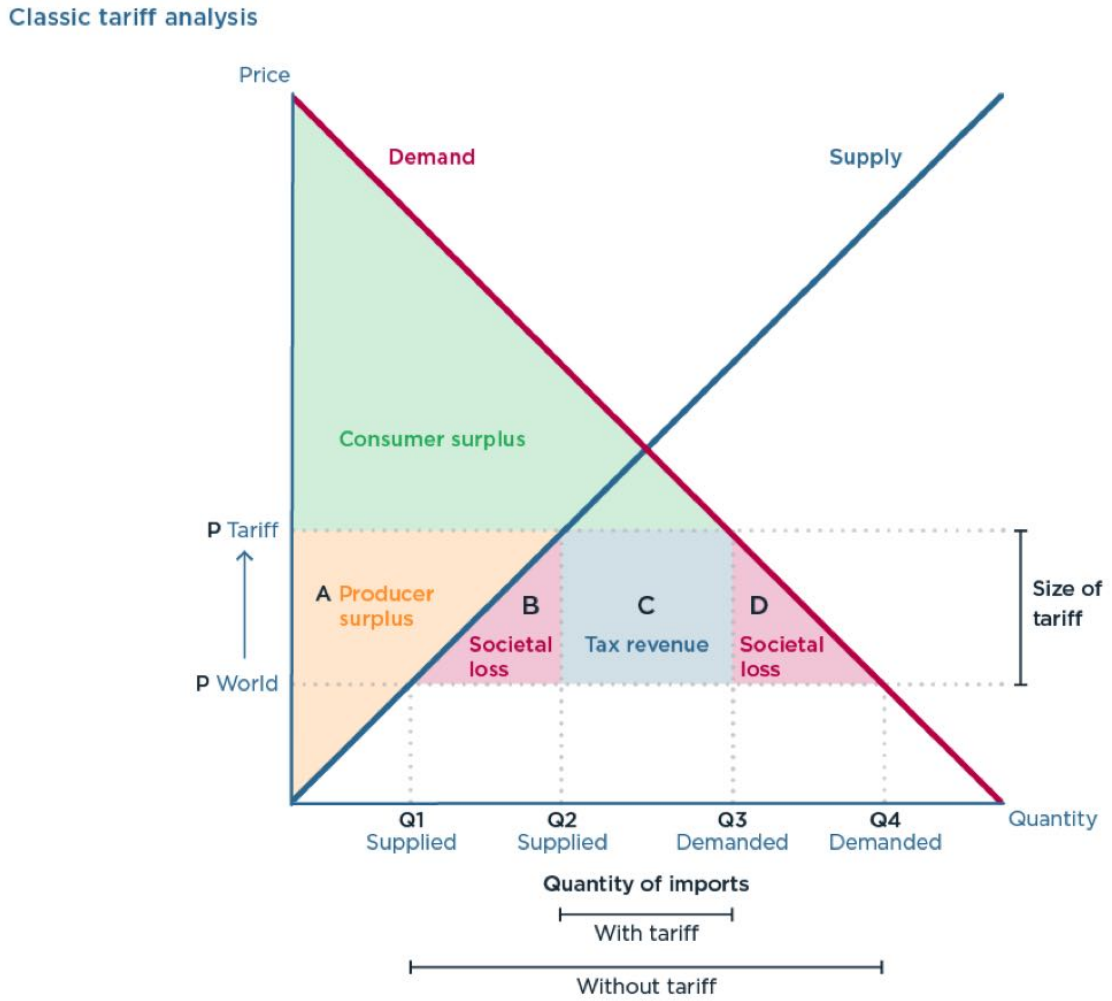


What If We Choose the “Small-Pie” Policy?



Suppose: $W = U_{Consumers} + U_{Producers}$

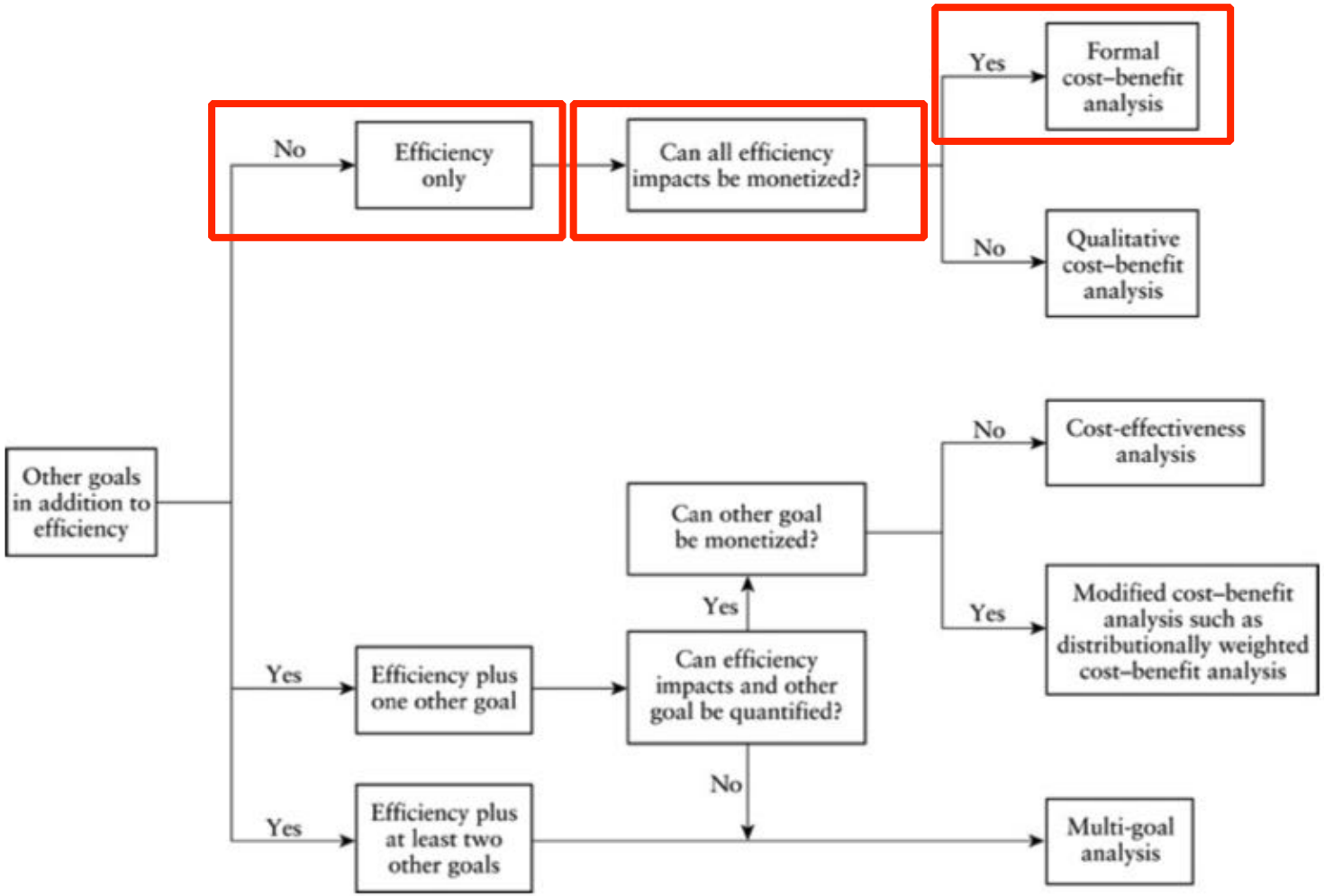
Analyse the welfare impact of tariffs in the U.S.!



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CBA: Identifying Potential Pareto Improvements in Practice

"Is policy project P efficient?"



Weimer & Vining 2017, Ch. 15

- What policymakers need: a response based on "actionable," $-quantifiable evidence.
- Cost-benefit analysis, generic form:

  Net Present Value (NPV) of P =

  present value of P's **Benefits**

  – present value of P's **Costs**.
- If the benefits flowing from a proposed project P exceed the costs occasioned by P, P should be undertaken!
- A positive NPV indicates a potential for "growing the utility pie" *and* a potential that losers be compensated.

Example: NPV of Non-Pharmaceutical Interventions (NPIs)

Value of a statistical life

Deaths without & with social distancing

Planning horizon

GDP forecast with & without SD

Social discount rate

$$\text{NPV} = \text{VSL} \times (D_1 - D_2) - \sum_{t=0}^{T} (Y_{1t} - Y_{2t})(1+r)^{-t}$$

Benefits (lives saved)

Costs (present value of GDP lost)

From Linda Thunström et al. 2020. "The Benefits and Costs of Using Social Distancing to Flatten the Curve for COVID-19." *Journal of Benefit-Cost Analysis* 11(2), 179-195

Example: NPV of Non-Pharmaceutical Interventions (NPIs)

NPV of a policy mix reducing the contact rate within the U.S. population in Spring 2020 by 38%.

Table 1 Benchmark outcomes for the uncontrolled scenario (without social distancing) and controlled scenario (with social distancing).

	Uncontrolled	Controlled
Infections (million)	287	188
Deaths (million)	2.18	0.941
Present value of GDP loss (trillion US$)	6.49	13.7
Value of lives lost (trillion US$)	21.8	9.41
Net benefits (trillion US$)		5.16

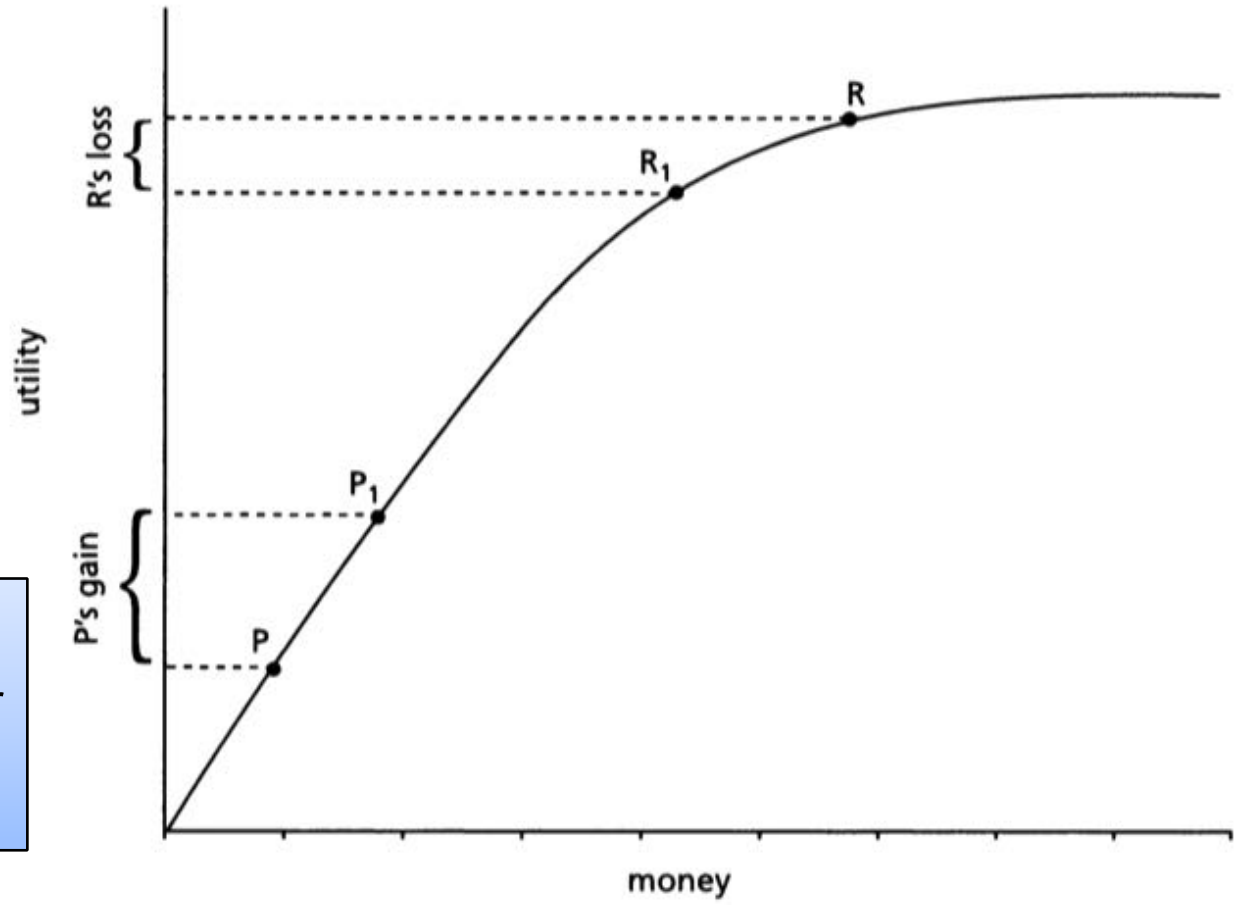
Benchmark parameter values: $R_0 = 2.4$, infectious period = 6.5 days, low mortality rate = 0.5 %, high mortality rate = 1.5 %, reduction in contact rate = 38 %, VSL = $10 million, uncontrolled initial GDP decline = 2.0 %, controlled initial GDP decline = 6.2 %, medical capacity threshold = 36 million infected, equal proportional rates of recovery in uncontrolled and controlled scenario, discount rate = 3.0 % $year^{-1}$, planning horizon = 30 years.

But The Separation Between Utility and Money May Fall Apart!

If a Euro is worth more in utility to the winners than to the losers of a policy, a policy change can raise total utility while the winners' willingness to pay (WTP) is smaller than the losers' willingness to accept (WTA).

Compensation is impossible!

Assuming diminishing marginal utility of income, wealthier people typically have higher WTP/WTA for a given utility change than poorer people do.



Types of Market Failure

Example: Overprescribing Antibiotics



"In a way it would be better for the community that so many people would not take antibiotics, but I have a feeling that for the individual it is better for him or for her to take antibiotics. So here is a little bit of conflict of interest in a way...now antibiotics are cheap and no harm is done if antibiotics are prescribed once or twice a year for an upper respiratory tract infection or a little bronchitis. Now why should I deprive my patients?"

— US physician, interviewed in Butler et al. 1998, p. 639

Externalities

Externalities arise whenever some person's actions affect another person's well-being outside the market mechanism.

Externality	Not an externality
Doctors overprescribe antibiotics, which leads to increasing antibiotic resistence.	Doctors overprescribe antibiotics, which bids up medicament prices.

"In a way it would be better for the community that so many people would not take antibiotics, but I have a feeling that for the individual it is better for him or for her to take antibiotics. So here is a little bit of conflict of interest in a way...now antibiotics are cheap and no harm is done if antibiotics are prescribed once or twice a year for an upper respiratory tract infection or a little bronchitis. Now why should I deprive my patients?"

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Why don't people internalise their externalities?

Externalities as a Social Dilemma

“Prisoners’ Dilemma”

	Prescribe responsibly	Overprescribe
Prescribe responsibly	2,2	-1,3
Overprescribe	3,-1	0*,0*

The best response is always defection, no matter what the other doctor does!

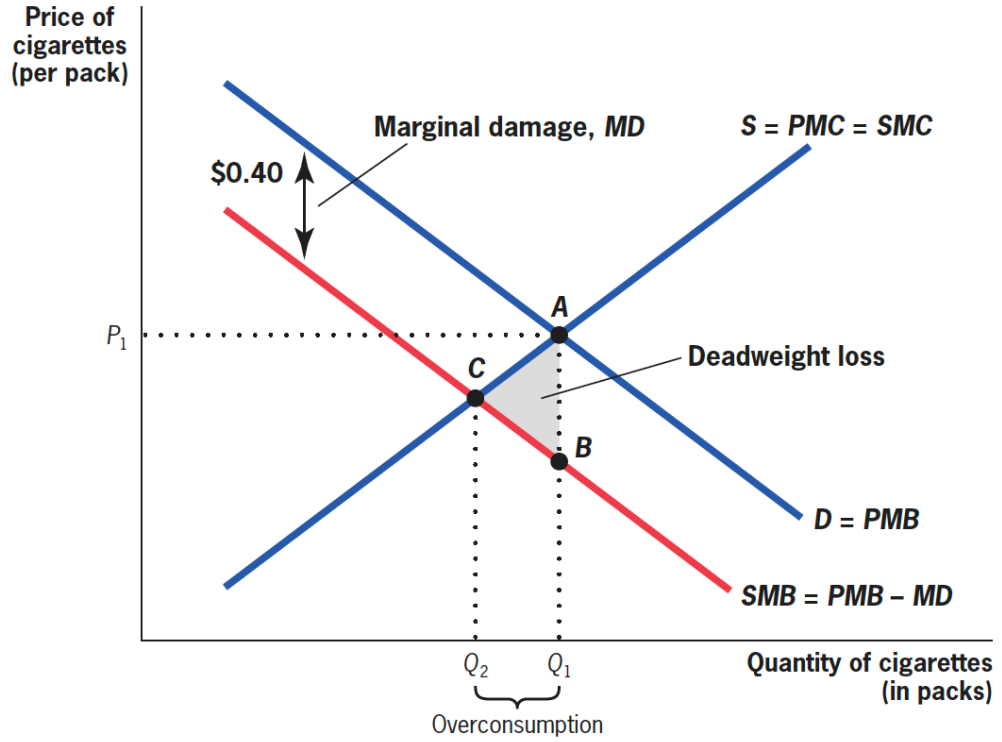
Failure to internalise externalities **shrinks the utility pie**!

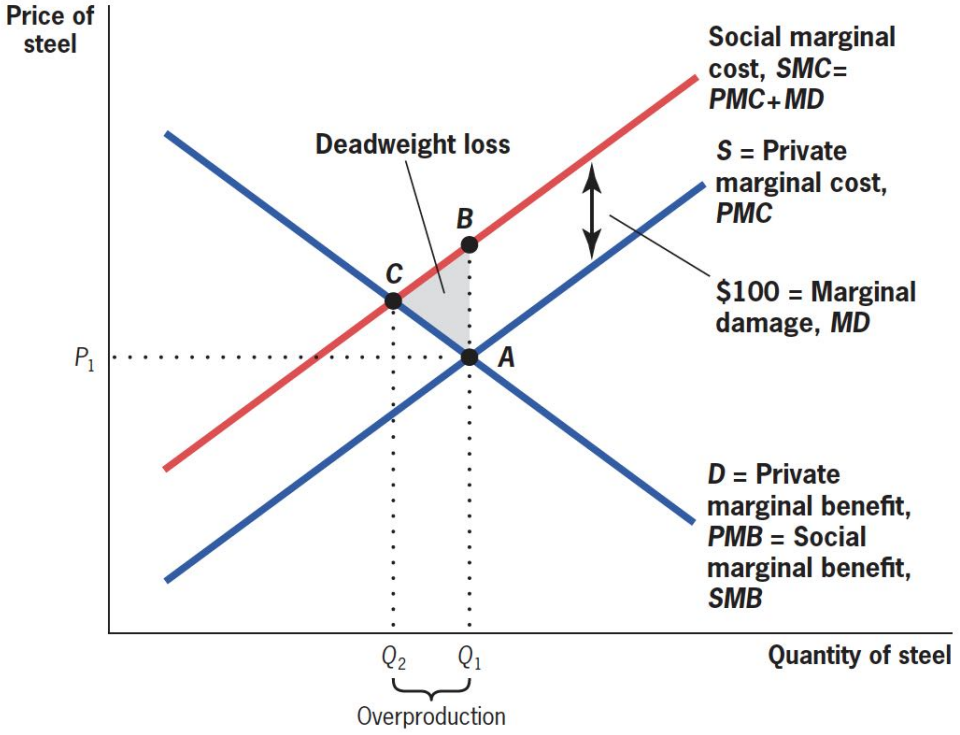
- Actions that impose negative externalities are done too much.
- Actions that impose positive externalities are done too little.

Negative Externalities

…lead to overproduction and overconsumption

- Negative **production** externalities: When a firm's production reduces the well-being of others who are not compensated by the firm.
- Negative **consumption** externalities: When an individual's consumption reduces the well-being of others who are not compensated by that individual.





Positive Externalities

…lead to underproduction and underconsumption

- Positive **production** externalities: When production increases the well-being of others but the firm is not compensated by those others.
  - Example: First-moving oil-drillers have a positive impact on the profit margins of competitors.
- Positive **consumption** externalities: When an individual's consumption increases the well-being of others but the individual is not compensated by those others.
  - Example: Vaccines

Collective Action

Collective Action:

Coordinated group action designed to achieve a common goal that individuals acting on their own could not otherwise obtain.

Consider a group of N > 1 people, sharing a common goal. Contribution cost is c > 0.

Whether or not (s)he contributes, the individual will benefit B > c.

The probability that the goal is achieved is $\frac{n}{N}$

Suppose the individual believes that *n* other people will contribute. His/her best response is to contribute *iff*:

$$\overbrace{\left[\frac{n+1}{N}-\frac{n}{N}\right]B}^{\text{Incremental Benefit}} \geq \overbrace{c}^{\text{Incremental Cost}}.$$

What are the chances that the common goal will be achieved?

The “Free Rider Problem”

- Theorised by Mancur Olson in 1965 in *Logic of Collective Action.*
- The co-occurrence of concentrated costs and dispersed benefits keeps groups from furthering their own interests!
- Rational people fail to internalise the positive externalities from collective action. This results in “small-pie” Pareto inefficient outcomes!
- The free rider problem appears to be predominantly a problem of large groups.
- And yet: some groups successfully overcome the dilemma!

The Public Goods Typology

Non-excludability:

My having access to the good means you have access to it as well.

Non-rivalry:

My using the good does not reduce your access to the good. (= Zero marginal social costs of consumption, but positive marginal costs of production)

Congestability:

My using (a certain quantity) of the good exhibits negative externalities of consumption.

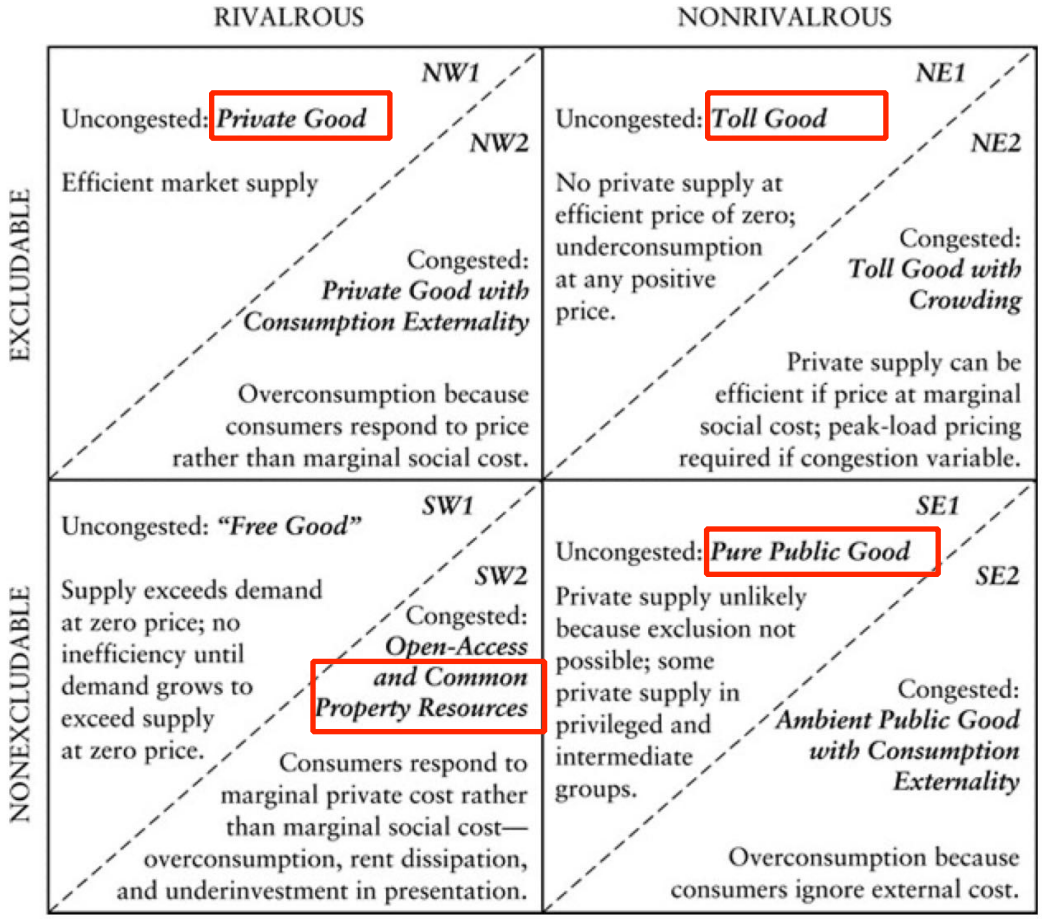
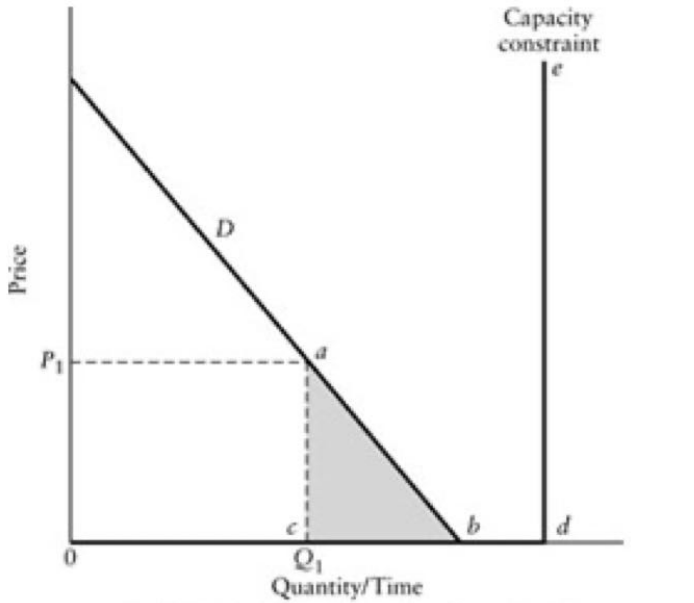


Figure 5.2 A Classification of Goods: Private and Public

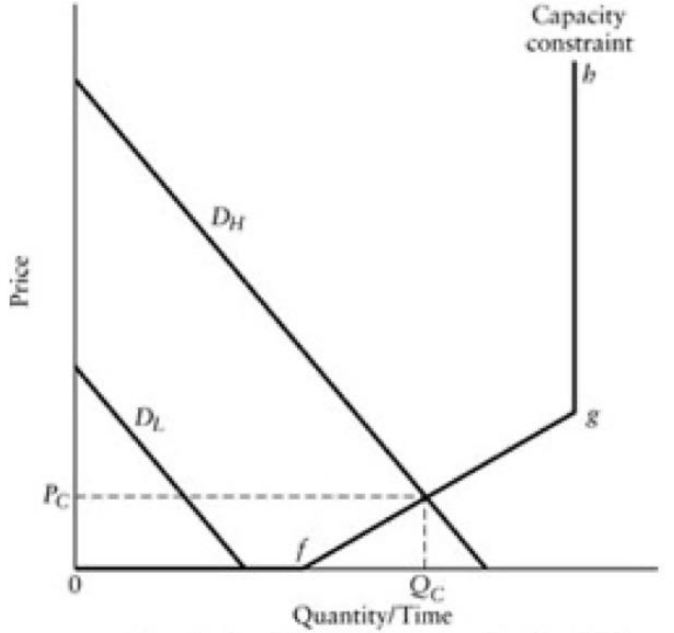
Toll Goods

- Excludable ownership, nonrivalrous in consumption.
- Bridges, tunnels, downtown streets, internet bandwith…
- If no congestion (MSCC = 0): any price P will create a Pareto inefficiency.
- If congestion (MSCC > 0 after some Q), Pareto efficiency can be achieved at P_C.
- See “congestion charge” in Stockholm, London.



Marginal social cost of consumption: line 0*de*

(a) Social Surplus Loss with Positive Price in Absence of Congestic

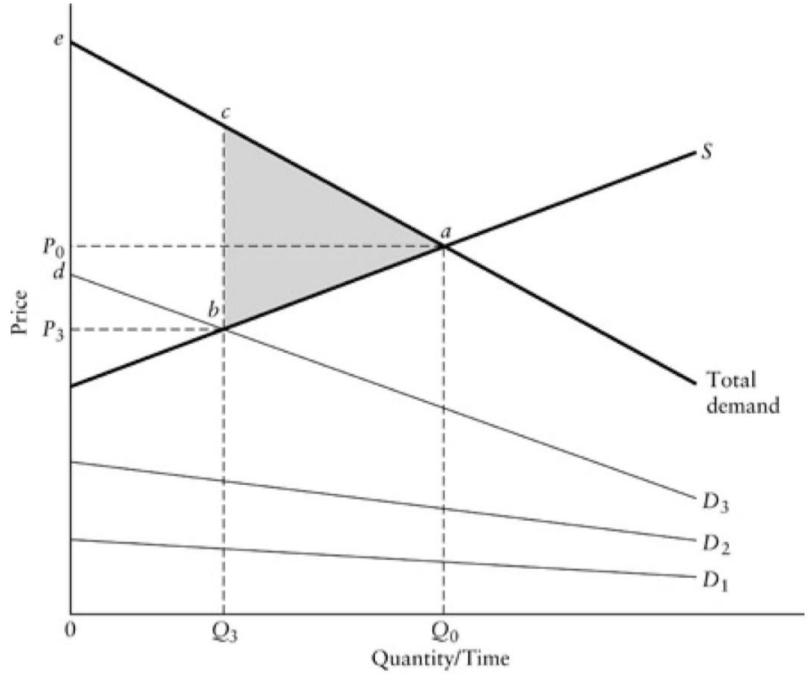


Marginal social cost of consumption: line 0*fgh*

(b) Appropriate Positive Price in Presence of Congestion

Pure Public Goods

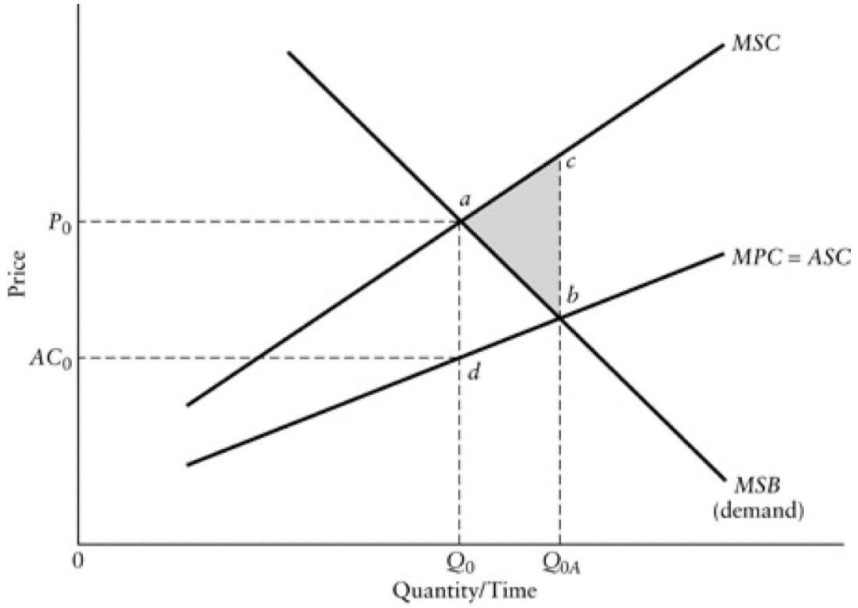
- Non-excludable ownership, nonrivalrous in consumption.
- Military defence, lighthouses, Wikipedia, Public broadcast, (some aspects of) the Internet.
- Affected to an extreme extent by collective action problems: impossibility to exclude free-riders will prevent these goods from being supplied by markets. Hence: Pareto inefficiency!
- Special (and rare) case of “privileged group” private provision.



Private Provision of a Public Good: Privileged Group

Open-Access (or Common Pool) Goods

- Non-excludable ownership, rivalrous in consumption.
- Fish stock, irrigation systems, oil, grassland, internet backbone, etc.
- "Tragedy of the Commons:" Nonexcludability leads to Pareto inefficient overconsumption and/or underinvestment — and ultimately — depletion of the good.
- Scarcity rent: If Q is limited at the Pareto-efficient level, who gains, who loses?

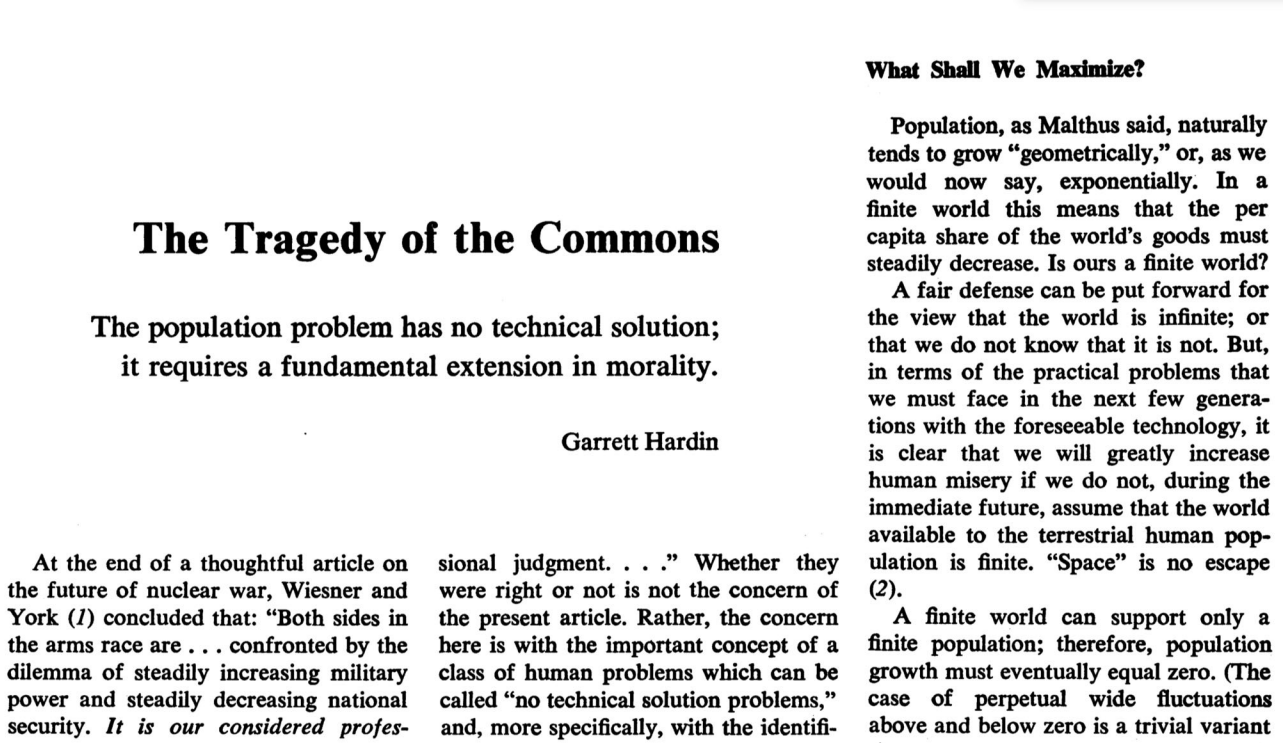


MPC = Marginal private cost
MSC = Marginal social cost
ASC = Average social cost

Socially optimal level of consumption: Q_0 (*MSB* = *MSC*)
Open access resource level of consumption: Q_{0A}
Deadweight loss from overconsumption: *abc*

Overconsumption of Open-Access Resources

The “Tragedy of the Commons”



The Tragedy of the Commons

The population problem has no technical solution; it requires a fundamental extension in morality.

Garrett Hardin

At the end of a thoughtful article on the future of nuclear war, Wiesner and York (*1*) concluded that: “Both sides in the arms race are . . . confronted by the dilemma of steadily increasing military power and steadily decreasing national security. *It is our considered professional judgment. . . .*” Whether they were right or not is not the concern of the present article. Rather, the concern here is with the important concept of a class of human problems which can be called “no technical solution problems,” and, more specifically, with the identifi-

What Shall We Maximize?

Population, as Malthus said, naturally tends to grow “geometrically,” or, as we would now say, exponentially. In a finite world this means that the per capita share of the world’s goods must steadily decrease. Is ours a finite world?

A fair defense can be put forward for the view that the world is infinite; or that we do not know that it is not. But, in terms of the practical problems that we must face in the next few generations with the foreseeable technology, it is clear that we will greatly increase human misery if we do not, during the immediate future, assume that the world available to the terrestrial human population is finite. “Space” is no escape (*2*).

A finite world can support only a finite population; therefore, population growth must eventually equal zero. (The case of perpetual wide fluctuations above and below zero is a trivial variant

	50 Cows	100 Cows
50 Cows	5,5	3,6
100 Cows	6,3	4*,4*

Garrett Hardin (1968) “The Tragedy of the Commons.” *Science* 162(3859), pp. 1243-1248.

Implications of Market Failures for Public Policy

How to Get People to Internalise their Externalities?

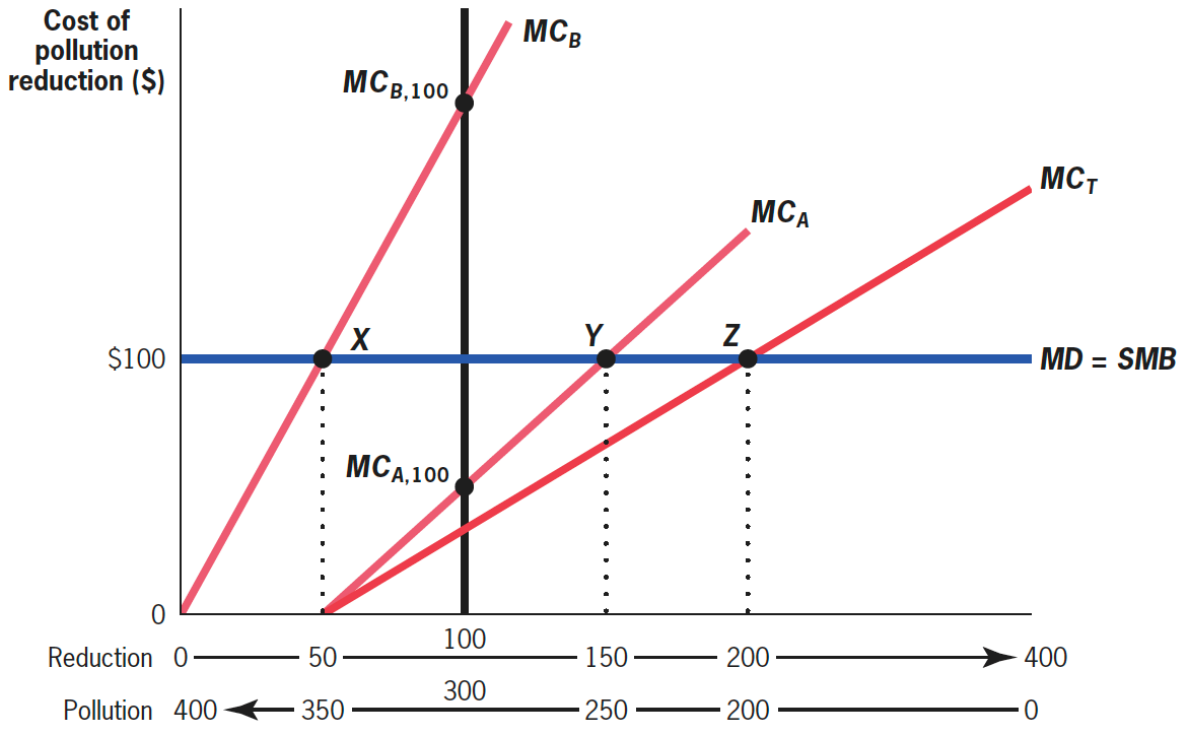
1. The force of persuasion.
2. Pigouvian subsidies and taxes
3. Direct regulation
4. Coasian bargaining
5. Decentralised self-governance

The Force of Persuasion

- Why can’t policymakers simply explain to people that internalising their externalities would make everyone better off?
- Remember the Prisoners’ Dilemma: even if I think that not everybody in my society is a selfish crook, it is still a best response for me to shirk.
- Even if everyone else is chipping in, it is still not in my interest to join…
- Some more intrusive kind of policy action is called for!

Tax vs. Regulatory Solutions to Externalities

- **Price**-based instruments:
  - Punitive (Pigouvian) tax equal to the negative externality (MD).
  - Generally efficient (second-best).
- **Direct regulation** without trade:
  - Government forces firms to produce the socially efficient quantity.
  - Generally not efficient.
- **Direct regulation** with trade ("cap and trade" schemes):
  - Government issues permits that allow a certain amount of pollution and let the firms trade.
  - Efficient, even if MC uncertain!



Second-best policy:
The policy that maximises social welfare, taking into consideration all the various effects of the policy.

A Market in Externalities

The “Coase Theorem”

- Are externalities really outside the market mechanism?
- The “Coase Theorem:”
  - *The Problem of Social Cost* (1960)
  - Part I: The party causing the externality and the party affected by the externality will voluntarily bargain about any issue over which they conflict.
  - Part II: The outcome of the bargaining between the parties will be Pareto-efficient, regardless of the initial distribution of property rights.
- George J. Stigler (NP 1982): “How so fine an economist could make such an obvious mistake.”



Ronald H. Coase 1910-2013

Implications of the Coase Theorem

- Coase (1960) makes two crucial assumptions:
  - **Property rights are clearly assigned:** economic agents exactly know who has the right to pollute and who not.
  - **Transaction costs are sufficiently low:** costs of identifying stakeholders, negotiating contracts, monitoring for compliance, sanction mechanisms…
- Conclusion of the practitioner:



- Conclusion of the economist: the market mechanism informs institutional design!

Implications of the Coase Theorem

- Coasean bargaining might be an effective internalization strategy in the presence of small-scale, localized externalities.
- Unsuitable for large-scaled externalities (global warming etc.).
- Coase hope was that governments…
  - …improve the quality of their institutions in view of lowering transaction costs.
  - …create opportunities for the creation of new markets for trading externalities.
  - Better define the role of the government in solving externality problems.

Problems With Coasean Bargaining in the Real World

1. Unruly Passenger 1 (claimed right to recline seat)
2. Unruly Passenger 2 (claimed right to legroom)

Post Nation

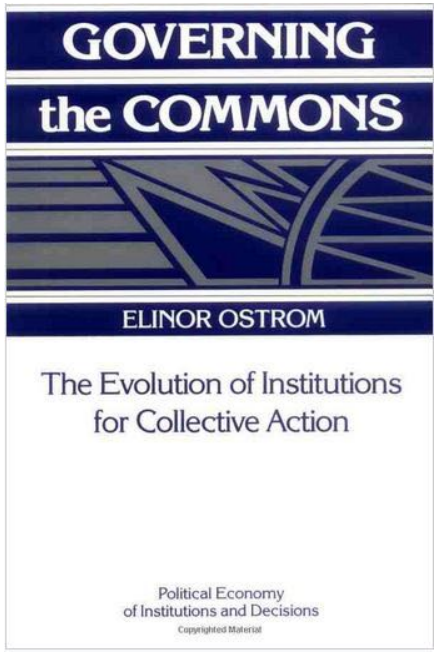
United flight diverted after passengers fight over legroom

By Abby Phillip August 26, 2014



Self-Governance

- The “Tragedy of Commons” is neither universal nor unavoidable!
- Collective Self-Governance can overcome the “Tragedy”
- State ownership and privatization are not the only solutions to the free rider problem
- Hardin’s “Tragedy” happens when
  - resource stakeholders are anonymous
  - get no feedback on individual actions
  - do/can not communicate



Ostrom, Elinor. 1990. *Governing the Commons: The Evolutions of Institutions for Collective Action*. Cambridge UP.



Elinor Ostrom (1933-2012)

Self-Governance: Ostrom's Eight Design Principles

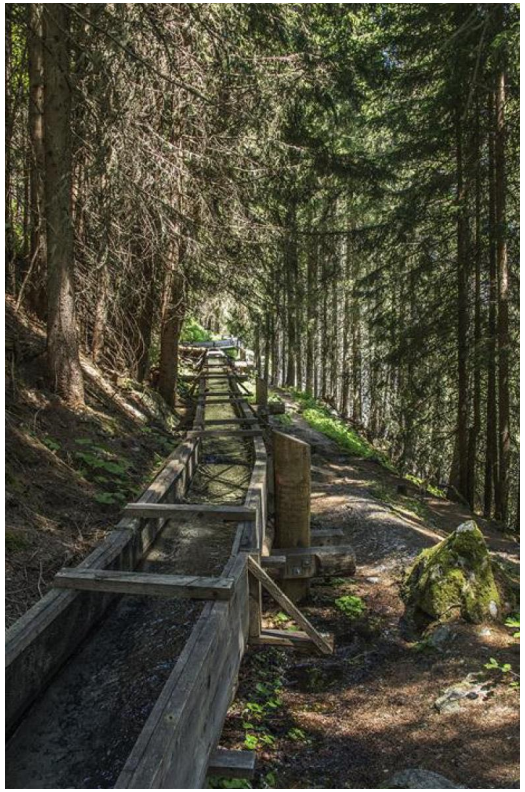
Ostrom's institutional design principles	Description
1. Clear boundaries	Individuals who have the right to use the resource are clearly defined
2. Congruence with local conditions	Rules restricting time, place, technology, & quantity of resource use are adapted to local conditions
3. Collective-choice arrangements	Individuals affected by resource-related rules can participate in modifying them
4. Monitoring	Resource conditions & use are monitored by the users themselves or by people accountable to users
5. Graduated sanctions	Credible threats & enforcement of penalties exist for users who violate resource-related rules
6. Conflict-resolution mechanisms	Users & officials have rapid & low-cost access to mechanisms of resolving conflicts
7. Recognition of right to organization	The rights of users to devise their own institutions are not challenged by government authorities
8. Multi-level governance	Appropriation, provision, monitoring, conflict-resolution are organised in multiple, nested layers

Successful Common Pool Self-Governance over Centuries: Land and Irrigation Regimes





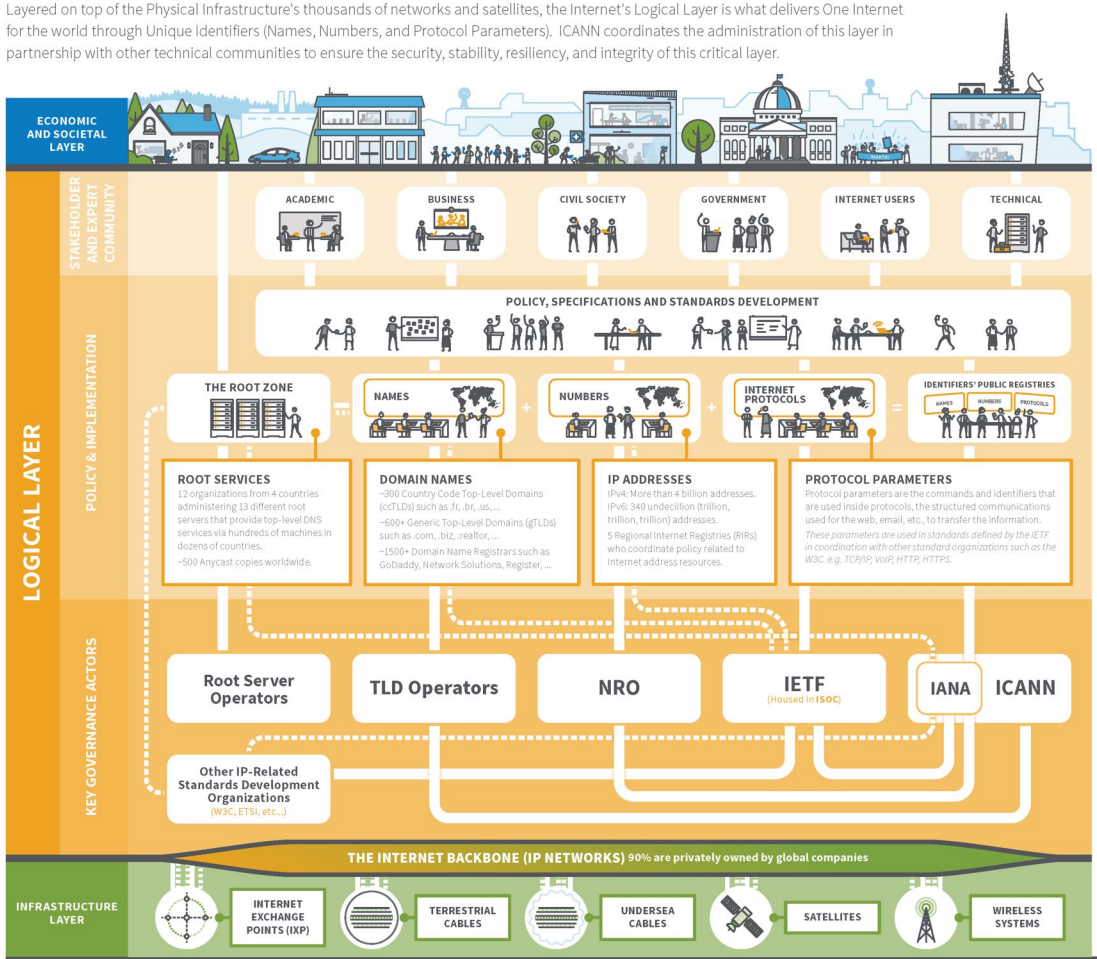




The Global Governance of the Internet

THE LOGICAL LAYER OF DIGITAL GOVERNANCE

Layered on top of the Physical Infrastructure's thousands of networks and satellites, the Internet's Logical Layer is what delivers One Internet for the world through Unique Identifiers (Names, Numbers, and Protocol Parameters). ICANN coordinates the administration of this layer in partnership with other technical communities to ensure the security, stability, resiliency, and integrity of this critical layer.



TECHNICAL OPERATIONS

The technical operating community is made up of multiple independent actors bound by common principles and mutual commitments that ensure the security and stability of the Internet Infrastructure. Each actor's community develops policies and standards in an open, inclusive, and consensus-based approach.

KEY GOVERNANCE ACTORS

ICANN *Internet Corporation for Assigned Names and Numbers*
Helps coordinate the Internet's systems of unique identifiers including domain names and IP addresses, as well as manages the IETF's protocol parameter registries.
www.icann.org

IANA, the Internet Assigned Numbers Authority, is a set of functions housed and operated within ICANN. It acts as the top-level allocator for blocks of IP addresses and AS numbers, proposes creation of and changes to DNS top-level domains, and manages lists of unique identifiers used in Internet protocols.
www.iana.org

IETF *Internet Engineering Task Force*
Develops and promotes a wide range of Internet standards dealing in particular with standards of the Internet protocol suite. Their technical documents influence the way people design, use, and manage the Internet. The IETF operates under the Internet Society (ISOC) with architectural oversight provided by the Internet Architecture Board (IAB).
www.ietf.org

ISO *International Organization for Standardization*
Standardizes, among many other things, the official names and postal codes of countries, dependent territories, special areas of geographic significance.
www.iso.org

NRO *Number Resource Organization*
A coordinating body for the five Regional Internet Registries (RIRs). The RIRs manage the distribution of IP addresses and Autonomous System Numbers in their regions of the world.
www.nro.net
AFRINIC *www.afrinic.net*
APNIC *www.apnic.net*
ARIN *www.arin.net*
LACNIC *www.lacnic.net*
RIPE NCC *www.ripe.net*

TLD Operators *Top Level Domain Operators*
Organizations which have been assigned the management of Top-Level Domains such as: Generic TLDs (.com, .edu, .info, .name etc ...), Country Code TLDs (.fr, .us, .gh, .cn etc...) and non-ASCII alphabet TLDs (in language such as Chinese, Korean, Arabic, Russian, French etc...) —among others.

Root Server Operators
12 independent organisations operate the 13 authoritative name servers (A through M) that serve the Domain Name System (DNS) root zone. The name servers are a network of hundreds of physical servers located in many countries around the world.
www.root-servers.org

W3C
The World Wide Web Consortium (W3C) is an international community where Member organizations, a full-time staff, and the public work together to develop Web standards. W3C's mission is to lead the Web to its full potential.
www.w3.org

STAKEHOLDER AND EXPERT COMMUNITY

Academic
- Institutions of higher learning
- Academic thought leaders
- Professors & students

Business
- Private-sector companies from across industries
- Industry and trade associations

Civil Society
- International organizations
- Non-governmental organizations
- Non-profit organizations
- Think Tanks

Government
- National governments
- Distinct economies recognized in international fora
- Multinational governmental and treaty organizations
- Intergovernmental organizations
- Public authorities (with a direct interest in global Internet Governance)

Internet Users
- Private citizens interested in regional or global Internet Governance

Technical
- Internet engineers
- Computer engineers
- Software developers
- Network operators

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